CS471 – Web Technologies (Laboratory)		Lab 6 — HTML Form

This lab session covers the basics of HTML form. By the end of this session, students should have a very good understanding of how to build a basic form and how to Handle form in Django.

Pre-lab Preparation:

1. Having bookmodule and usermodule in apps directory, each containing separate URL configurations.
2. Template directory located in 'apps' with necessary settings.
3. Master template 'base.html'

Lab Activities: A complete webpage example with html form and Handling form in Django

Task 1: Build HTML form template

Step 1: Using the master template base.html from Lab4, import it in <head> section in base.html the following:

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">
```

```
    <link rel="stylesheet" href="{% static 'styles.css' %}">
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">
    {% block stylesheets %}{% endblock stylesheets %}
</head>
<body>

    {% include "includes/header.html" %}

```

Step 2: using the master template 'base.html', Create a view function and URL '/books/search' to a page that contains the following HTML form:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} Searching Page {% endblock title %}
{% block stylesheets %}
<style>
    .searchForm{
        margin-top: 20px;
    }

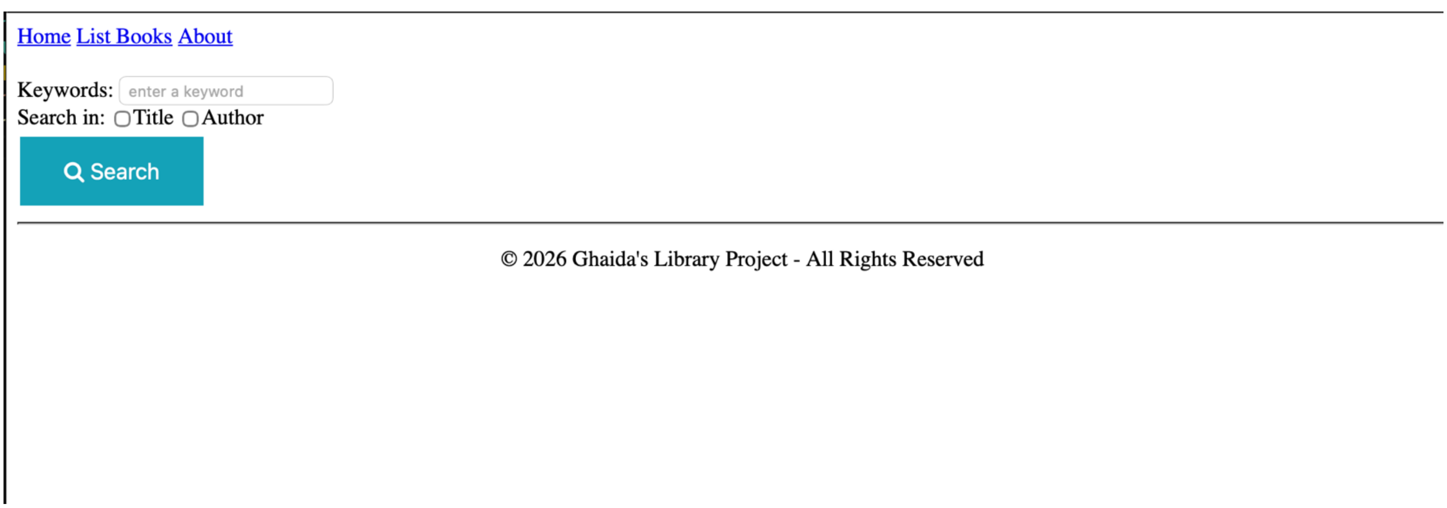
    button {
        color: #fff;
        background-color: #17a2b8;
        border-color: #17a2b8;
        border: none;
        padding: 16px 32px;
        text-align: center;
        text-decoration: none;
        display: inline-block;
```

```
        font-size: 16px;
        margin: 4px 2px;
        transition-duration: 0.4s;
        cursor: pointer; /* hand pointer */
    }
    button:hover {
        background-color: green;
        color: white;
    }
</style>
{% endblock stylesheets %}

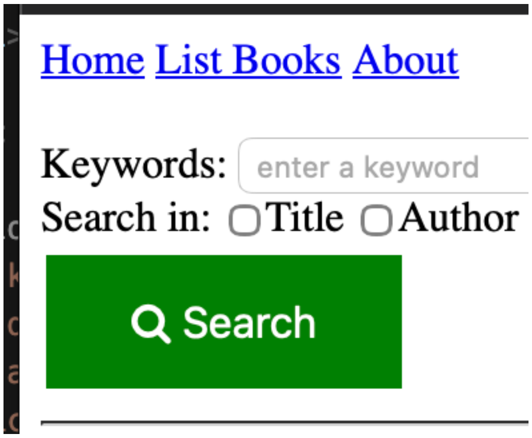
{% block content %}
<form class="searchForm" method="post" action="">
    {% csrf_token %}
    <label>Keywords:</label>
    <input type="text" name="keyword" placeholder="enter a keyword"/>
    <br/>

    <label>Search in:</label>
    <input type="checkbox" name="option1"/><label>Title</label>
    <input type="checkbox" name="option2"/><label>Author</label>
    <br/>

    <button type="submit" ><i class="fa fa-search" aria-hidden="true"></i> Search</button>
</form>
{% endblock content %}
```



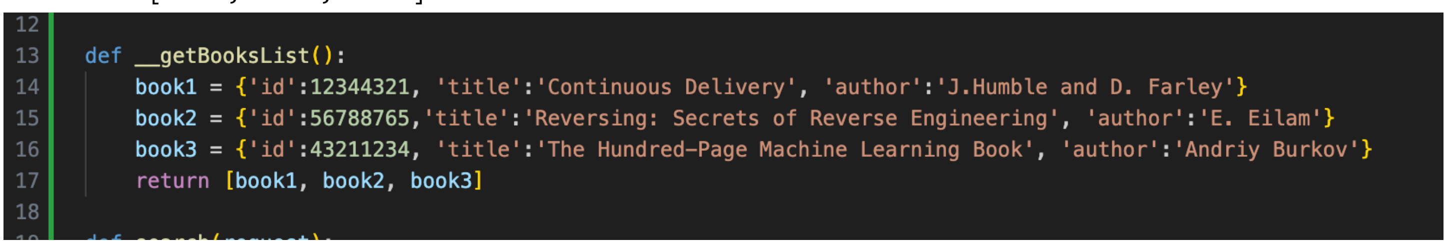
Step 3: After running the server, open the web browser and navigate to http://127.0.0.1:8000/books/search. You should see the form.



Task 2: Handle the form once submitted

Step 1: In 'apps/bookmodule/view.py', create the following function:

```
def __getBooksList():
    book1 = {'id':12344321, 'title':'Continuous Delivery', 'author':'J.Humble and D. Farley'}
    book2 = {'id':56788765,'title':'Reversing: Secrets of Reverse Engineering', 'author':'E. Eilam'}
    book3 = {'id':43211234, 'title':'The Hundred-Page Machine Learning Book', 'author':'Andriy Burkov'}
    return [book1, book2, book3]
```



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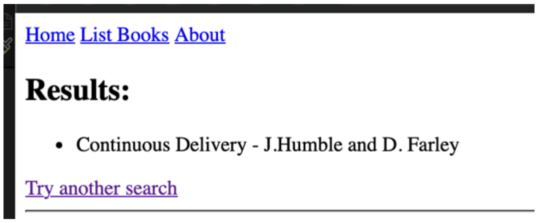
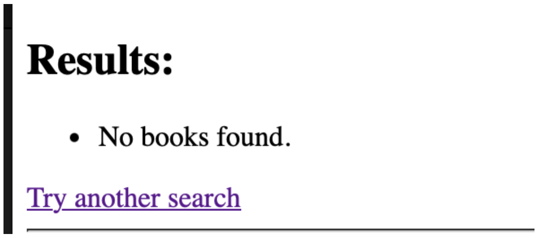
```

Step 2: In the same corresponding view/controller in views.py that shows the form, add the code where If there is an action in the form, the form needs to be handled in that view.

Note: it is possible to direct the action to an exclusive view function specific to handle the submitted form.

```
if request.method == "POST":
        string = request.POST.get('keyword').lower()
        isTitle = request.POST.get('option1')
        isAuthor = request.POST.get('option2')
        # now filter
        books = __getBooksList()
        newBooks = []
        for item in books:
            contained = False
            if isTitle and string in item['title'].lower(): contained = True
            if not contained and isAuthor and string in item['author'].lower():contained = True
            if contained: newBooks.append(item)
        return render(request, 'bookmodule/bookList.html', {'books':newBooks})
```





Step 3: Create an HTMl file '`bookList.html`' to list the books (you should use the master template '`base.html`')√

Step 4: Now from the web browser, navigate to http://127.0.0.1:8000/books/search, and submit the form with data.